

Tutorial Sheet - III

Q.1. Let $C([0,1]) = \{f: [0,1] \rightarrow \mathbb{C} : f \text{ is continuous}\}$.

Define $T: C([0,1]) \rightarrow \mathbb{C}$ by

$$T(f) = \int_0^1 f(x)x^3 dx, \quad \forall f \in C([0,1]).$$

Show that T is a linear map.

Q2. Let X be a vector space over F . If $x \in X$ is a non-zero vector, show that there exists a linear map $T: X \rightarrow F$ such that

$$T(x) \neq 0.$$

Q3. Let $C([-1,1]) = \{f: [-1,1] \rightarrow \mathbb{R}, f \text{ is continuous}\}$.

Define $P: C([-1,1]) \rightarrow C([-1,1])$ by

$$(Pf)(x) = \frac{f(x) + f(-x)}{2}, \quad \forall x \in [-1,1] \text{ \& } f \in C([-1,1]).$$

Show that P is a linear map and P is a projection, that is $P^2 = P$.

Q4. Let X be the vector space of polynomials with complex co-efficients of degree less than n .

Let $z_0, \dots, z_n$ be distinct complex numbers.

Define a map $T: X \rightarrow \mathbb{C}^{n+1}$ by

$$T(p) = (p(z_0), \dots, p(z_n)), \quad \forall p \in X.$$

(i) Show that T is a linear map.

(ii) The null space of T is trivial and T is an
a surjective map.

(iii) Show that there exist n numbers $m_1, \dots, m_n$ such that $\int_0^1 p(x) dx = m_1 p(t_1) + \dots + m_n p(t_n) \forall p \in X$ where $t_1, \dots, t_n$ are n distinct points in $[0, 1]$.

(iv) • Show that the map $T: X \rightarrow \mathbb{C}^n$ defined by $T(p) = (p(s_1), \dots, p(s_n))$ is surjective.

- For $(\alpha_1, \dots, \alpha_n) \in \mathbb{C}^n$, find a polynomial p in X such that $p(s_i) = \alpha_i, \forall i=1, \dots, n$.